Chapter 2. Kepler's Equation and Kepler's Problem

2.1. Historical Background

2.2. Kepler's Equation

- Kepler's Equation
  - Allows us to determine the relation between time displacements and angular displacements
  - Given that we know/determine the orbit $(a, e, i, \cdots)$
  - Critical interests in many applications for satellites, planets, etc.

- Derivation of Kepler's Equation

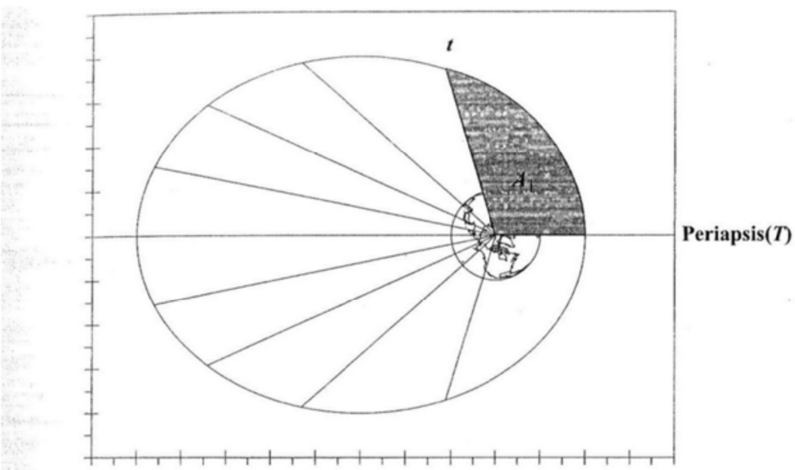


Figure 2-1. Kepler's Equation—Equal Areas. According to Kepler's second law, a satellite will sweep out equal areas in equal time. This figure shows positions for ten equal spacings in an orbit. For this particular orbit, a = 31,890 km (5.0 ER), e = 0.6, and the space between the lines is about 94.4 minutes.

Suppose that a satellite starts at periapsis at time T

Kepler's 2nd law of constant aerial velocity gives

$$\frac{t-T}{A_1} = \frac{P}{\pi ab} = \frac{P}{\pi a^2 \sqrt{1-e^2}} \tag{1}$$

Since we assume that we have determined the orbit, (i.e., the inertial position/velocity is known), the only unknown is A_1, which we determine by geometry:

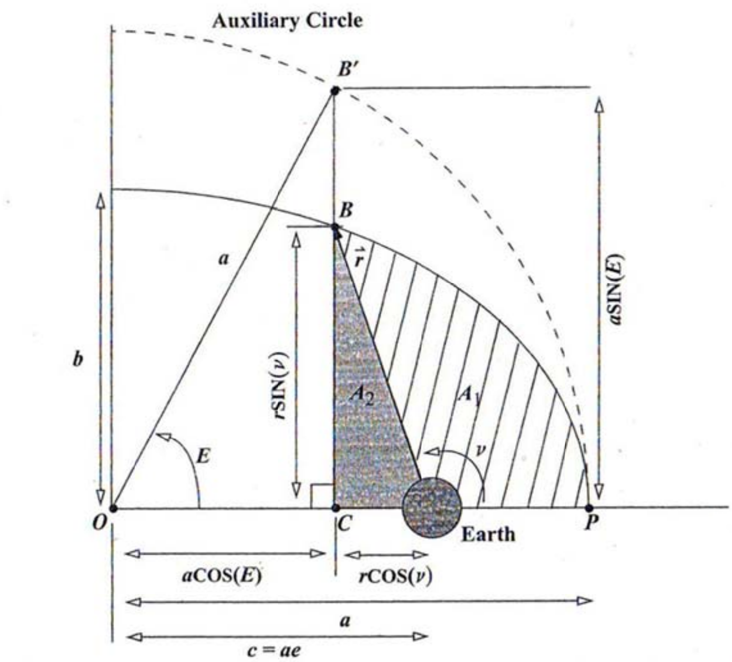


Figure 2-2. **Geometry of Kepler's Equation.** The eccentric anomaly uses an auxiliary circle as shown. The ultimate goal is to determine the area, A_1, which allows us to calculate the time.

We define an auxiliary circle and Eccentric Anomaly E and use E instead of ν to determine A_1

Recall

$$y_e = y_c\sqrt{1-e^2}$$
$$\Rightarrow BC = B'C\sqrt{1-e^2}$$

$$\Rightarrow BC = a\sin E\sqrt{1-e^2} \tag{2}$$

$$\begin{aligned}
A_1 &= PCB - A_2 \\
&= PCB'\sqrt{1-e^2} - A_2 \quad \text{(geometrical property of ellipse)} \\
&= (POB' - COB')\sqrt{1-e^2} - \frac{1}{2}(ae - a\cos E)BC \\
&= \left[\frac{1}{2}a^2E - \frac{1}{2}(a\cos E)(a\sin E)\right]\sqrt{1-e^2} - \frac{1}{2}(ae - a\cos E)a\sin E\sqrt{1-e^2} \\
&= \frac{1}{2}a^2\sqrt{1-e^2}(E - \sin E\cos E) - \frac{1}{2}a^2\sqrt{1-e^2}(e - \cos E)\sin E \\
&= \frac{1}{2}a^2\sqrt{1-e^2}(E - e\sin E)
\end{aligned}$$

Now from the 2nd law (1), we have

$$\frac{t-T}{A_1} = \frac{P}{\pi a^2\sqrt{1-e^2}}$$

$$\Rightarrow \frac{t-T}{\frac{1}{2}a^2\sqrt{1-e^2}(E - e\sin E)} = \frac{P}{\pi a^2\sqrt{1-e^2}}$$

$$\Rightarrow P = \frac{2\pi(t-T)}{E - e\sin E}$$

$$\left(\Uparrow \ P = 2\pi\sqrt{\frac{a^3}{\mu}} \quad \text{(Kepler's 3rd Law)}\right)$$

$$\Rightarrow \sqrt{\frac{a^3}{\mu}} = \frac{t-T}{E - e\sin E} \quad \text{(Kepler's Equation)}$$

Once you know the eccentric anomaly, you can determine the time of flight, or vice versa

$$\begin{cases} M \equiv E - e\sin E & \text{(Mean Anomaly)} \\ n \equiv \dfrac{2\pi}{P} = \sqrt{\dfrac{\mu}{a^3}} & \text{(Mean Motion)} \end{cases}$$

$$\Rightarrow M = n(t-T) = E - e\sin E$$

- $M = n(t-T)$ increases linearly as the orbit progresses
- n is the mean angular rate of the orbital motion

In general situations where the initial time is not located at periapsis

$$t - t_0 = (t - T) - (t_0 - T) = \frac{E - e\sin E}{n} - \frac{E_0 - e\sin E_0}{n}$$
$$\Rightarrow n(t - t_0) = (E - e\sin E) - (E_0 - e\sin E_0) = M - M_0$$
$$\Rightarrow M - M_0 = n(t - t_0) = (E - e\sin E) - (E_0 - e\sin E_0) + 2\pi k$$

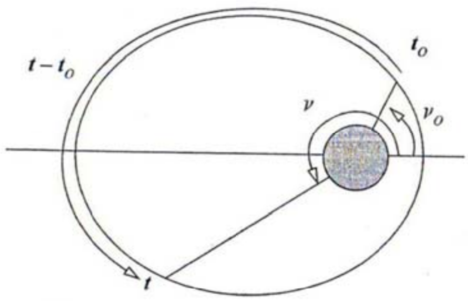


Figure 2-3. **General Time of Flight.** The time of flight between two specified locations has two basic forms: those of less than one period (as shown) and those with more than one period. Multiple periods include one or more complete revolutions in addition to the displacement shown here.

- Now that we have used E and M instead of ν, we now derive the relation between E and ν for an elliptic orbit, that is, for $e < 1$

- E from ν

Recall from Kepler's 1st law or trajectory equation

$$r = \frac{p}{1 + e\cos\nu}$$

$$\Rightarrow r = \frac{a(1 - e^2)}{1 + e\cos\nu} \qquad (1)$$

■ From the geometry of Kepler's equation

$$BC = \sqrt{1 - e^2}\, B'C$$
$$\Rightarrow r\sin\nu = \sqrt{1 - e^2}\,(a\sin E)$$

$$\Rightarrow \sin E = \frac{r \sin \nu}{a\sqrt{1-e^2}}$$

$$\Uparrow \quad (1) \quad r = \frac{a(1-e^2)}{1+e\cos\nu}$$

$$\Rightarrow \sin E = \frac{\sqrt{1-e^2}\,\sin\nu}{1+e\cos\nu} \qquad (2)$$

- Again, from the geometry of Kepler's equation

$$OC = OF - CF$$

$$\Rightarrow a\cos E = ae + r\cos\nu$$

$$\Rightarrow \cos E = \frac{ae + r\cos\nu}{a}$$

$$\Uparrow \quad (1) \quad r = \frac{a(1-e^2)}{1+e\cos\nu}$$

$$\Rightarrow \cos E = \frac{e+\cos\nu}{1+e\cos\nu} \qquad (3)$$

- Then,

$$\tan E = \frac{\sqrt{1-e^2}\,\sin\nu}{e+\cos\nu} \qquad (4)$$

- ν from E
  - Simply rearrange (3) for $\cos\nu$:

$$\cos\nu = \frac{\cos E - e}{1-e\cos E} \qquad (5)$$

  - Start by rearranging (2)

$$\sin\nu = \frac{\sin E(1+e\cos\nu)}{\sqrt{1-e^2}}$$

$$\Uparrow \quad (5) \quad \cos\nu = \frac{\cos E - e}{1-e\cos E}$$

$$\Rightarrow \sin\nu = \frac{\sin E\sqrt{1-e^2}}{1-e\cos E} \qquad (6)$$

■ Then,

$$\tan\nu = \frac{\sin E\sqrt{1-e^2}}{\cos E - e}$$

- Relation: $\tan\dfrac{\nu}{2} \Leftrightarrow \tan\dfrac{E}{2}$

  From the trigonometric identity,

$$\tan\frac{\nu}{2} = \frac{\sin\nu}{1+\cos\nu}$$

$$\Uparrow \quad (5)\ \cos\nu = \frac{\cos E - e}{1-e\cos E}, \quad (6)\ \sin\nu = \frac{\sin E\sqrt{1-e^2}}{1-e\cos E}$$

■

$$\Rightarrow \tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2}$$

$$\Rightarrow \tan\frac{E}{2} = \sqrt{\frac{1-e}{1+e}}\tan\frac{\nu}{2}$$

- Alternative form of trajectory equation (E instead of ν as independent variable)

$$r = \frac{p}{1+e\cos\nu} = \frac{a(1-e^2)}{1+e\cos\nu}$$

$$\Uparrow \quad (5)\ \cos\nu = \frac{\cos E - e}{1-e\cos E}$$

$$\Rightarrow r = a(1-e\cos E) \tag{7}$$

2.2.1. Alternative Formulation for *E*

- Derivation of Kepler's Equation by Analytical method (instead of Geometrical method)

  Start by

$$h = r^2 \dot{\nu} = \text{constant}$$

$$\Rightarrow h = r^2 \frac{d\nu}{dt}$$

$$\Rightarrow \int_T^t h dt = \int_0^\nu r^2 d\nu \tag{8}$$

Recall (5):

$$\cos\nu = \frac{\cos E - e}{1 - e\cos E}$$

$$\Rightarrow -\sin\nu d\nu = \frac{-\sin E(1-e^2)}{(1-e\cos E)^2} dE$$

$$\Rightarrow d\nu = \frac{\sin E(1-e^2)}{\sin\nu(1-e\cos E)^2} dE$$

$$\Uparrow \ (6) \ \ \sin\nu = \frac{\sin E\sqrt{1-e^2}}{1-e\cos E}$$

$$\Rightarrow d\nu = \frac{\sqrt{1-e^2}}{1-e\cos E} dE$$

$$\Uparrow \ (7) \ \ r = a(1-e\cos E)$$

$$\Rightarrow d\nu = \frac{a\sqrt{1-e^2}}{r} dE \tag{9}$$

From (8),

$$\int_T^t h dt = \int_0^\nu r^2 d\nu$$

$$\Uparrow \ (9) \ \ d\nu = \frac{a\sqrt{1-e^2}}{r} dE$$

$$\Rightarrow h(t-T) = \int_0^E ra\sqrt{1-e^2} dE$$

$$\Uparrow \ (7) \ \ r = a(1-e\cos E)$$

$$\Rightarrow h(t-T) = \int_0^E a^2\sqrt{1-e^2}(1-e\cos E) dE$$

$$\Rightarrow h(t-T) = a^2\sqrt{1-e^2}\,(E - e\sin E)$$

$$\Uparrow \;\; h = na^2\sqrt{1-e^2} \;\; \Leftarrow \begin{cases} h = \sqrt{\mu p} = \sqrt{\mu a(1-e^2)} \\ n = \sqrt{\dfrac{\mu}{a^3}} \end{cases}$$

$$\Rightarrow t - T = \sqrt{\frac{a^3}{\mu}}(E - e\sin E)$$

$$\Rightarrow M \equiv n(t-T) = E - e\sin E$$

2.2.2. Formulation for the Parabolic Anomaly

- For a complete analysis of Kepler's equation, we must derive expressions for parabolic/hyperbolic anomalies

- Deriving the equivalent of Kepler's equation for parabolic anomaly, we begin by

$$h = r^2\frac{d\nu}{dt} = \sqrt{\mu p} \tag{1}$$

Unlike the elliptic orbit case, we cannot define E because the semimajor axis is infinite

Consider trajectory equation:

$$r = \frac{p}{1 + e\cos\nu}$$

$$\Uparrow \;\; e = 1.0$$

$$\Rightarrow r = \frac{p}{2\cos^2\dfrac{\nu}{2}}$$

$$\Rightarrow r = \frac{p}{2}\frac{\sin^2\dfrac{\nu}{2} + \cos^2\dfrac{\nu}{2}}{\cos^2\dfrac{\nu}{2}}$$

$$\Rightarrow r = \frac{p}{2}\left(1 + \tan^2\frac{\nu}{2}\right) \tag{2}$$

Introducing (2) into (1) leads to

$$\sqrt{\mu p} = \frac{p^2}{4}\left(1+\tan^2\frac{\nu}{2}\right)^2 \frac{d\nu}{dt}$$

$$\Rightarrow \int_T^t 4\sqrt{\frac{\mu}{p^3}}dt = \int_0^\nu \left(1+\tan^2\frac{\nu}{2}\right)^2 d\nu$$

$$\Rightarrow 2\sqrt{\frac{\mu}{p^3}}(t-T) = \tan\frac{\nu}{2} + \frac{1}{3}\tan^3\frac{\nu}{2} \tag{3}$$

which is equivalent to Kepler's equation for a parabolic orbit

Defining the parabolic anomaly B and mean motion n_p as

$$B = \tan\frac{\nu}{2}$$

$$n_p = 2\sqrt{\frac{\mu}{p^3}}$$

we have

$$M = n_p(t-T) = B + \frac{1}{3}B^3,\ t(\nu=0) = T \tag{4}$$

Generalizing for arbitrary initial position, we have

$$\frac{M-M_0}{n_p} = t - t_0 = \frac{1}{2}\sqrt{\frac{p^3}{\mu}}\left[\left(B+\frac{1}{3}B^3\right) - \left(B_0 + \frac{1}{3}B_0^3\right)\right] \tag{5}$$

Note that we use '*p*' to scale the parabola because *a* becomes infinite for parabola

Also, introducing the definition of parabolic anomaly B into (2), we have

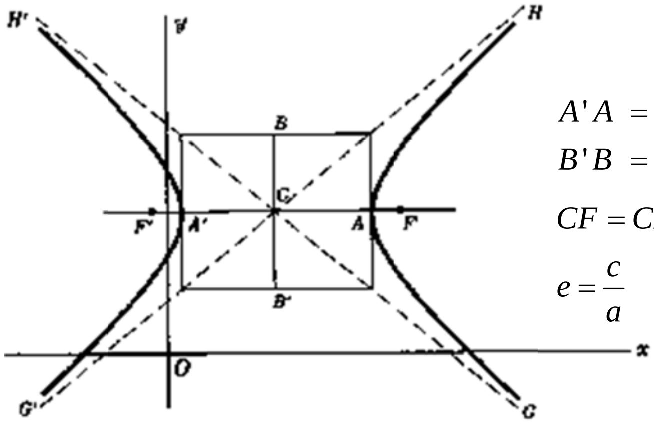
$$r = \frac{p}{2}\left(1+B^2\right)$$

which is a trajectory equation as a function of parabolic anomaly B

Mathematically, $-180 < \nu < 180$, but ν never reaches ± 180 for practical cases

2.2.3. Formulation for the Hyperbolic Anomaly

- Hyperbolic orbit occurs when studying the comet and the interplanetary/interlunar orbits



$$A'A = 2a$$
$$B'B = 2b$$
$$CF = CF' = c = \sqrt{a^2 + b^2}$$
$$e = \frac{c}{a}$$

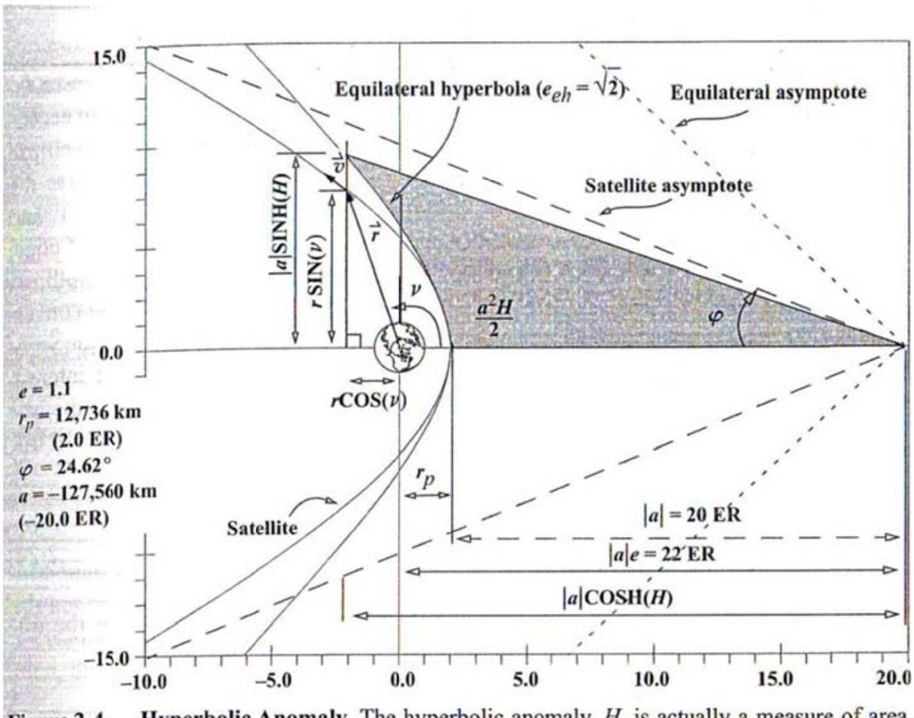


Figure 2-4. **Hyperbolic Anomaly.** The hyperbolic anomaly, H, is actually a measure of area, shown as the shaded region in the figure. Notice we use an equilateral hyperbola to determine H, just as we use a circle for elliptical orbits. Semimajor axis quantities are shown as absolute values to keep distances positive. The satellite and equilateral hyperbola have the same semimajor axis value. Thus, the focus for the equilateral hyperbola does not coincide with that of the satellite and its r_p = 8.28 ER.

- Characteristics of Hyperbola

  - $\xi = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} > 0 \;\; \Rightarrow \;\; a < 0$

  - Asymptotes represented by the turning angle

    $$\tan\varphi = \frac{b}{a} = \sqrt{e^2 - 1}$$

  - The trajectory equation leads to

    $$-180^\circ + \cos^{-1}\frac{1}{e} < \nu < 180^\circ - \cos^{-1}\frac{1}{e}$$

- We introduce an equilateral hyperbola with

  $$e_{eh} = \sqrt{2}$$

  Then, we have

  $y_h = y_{eh}\sqrt{e^2 - 1}$ (which is analogous to $y_e = y_c\sqrt{1 - e^2}$)

- The Hyperbolic Anomaly *H* is defined by using an auxiliary conic section, as we do for the Eccentric Anomaly *E*

- *H* also represents the area from the equilateral hyperbola.

- From the geometry, we obtain *H* from *ν*

- $$\begin{aligned} &|a|\cosh H = |a|e - r\cos v \qquad \Leftarrow |a| = -a \\ &\Rightarrow \cosh H = \frac{ae + r\cos(v)}{a} \qquad \Leftarrow r = \frac{p}{1+e\cos v} = \frac{a(1-e^2)}{1+e\cos v} \\ &\Rightarrow \cosh H = \frac{e+\cos v}{1+e\cos v} \end{aligned} \tag{1}$$

- $$\begin{aligned} &|a|\sinh H = \frac{-r\sin v}{\sqrt{e^2-1}} \qquad \Leftarrow r = \frac{a(1-e^2)}{1+e\cos v},\ |a| = -a \\ &\Rightarrow \sinh H = \frac{\sin v\ \sqrt{e^2-1}}{1+e\cos v} \end{aligned} \tag{2}$$

- Also, we can determine v from H
  - From (1),

    $$\cos v = \frac{\cosh H - e}{1 - e\cosh H} \tag{3}$$

  - From (2),

    $$\sin v = \frac{\sinh H(1+e\cos v)}{\sqrt{e^2-1}} \qquad \Leftarrow (3)$$

    $$\sin v = \frac{-\sin H\sqrt{e^2-1}}{1-e\cosh H} \tag{4}$$

  - $$r = \frac{a(1-e^2)}{1+e\cos v}$$

    $$\Rightarrow\ r\ = a(1-e\cosh H) \tag{5}$$

$$\tanh\frac{H}{2} \equiv \frac{\sinh H}{1+\cosh H}$$

- $$\Rightarrow \tanh\frac{H}{2} = \frac{\sin v\sqrt{e^2-1}}{(1+e)(1+\cos v)}$$

$$\Rightarrow \begin{cases} \tanh\dfrac{H}{2} = \sqrt{\dfrac{e-1}{e+1}}\tan\dfrac{v}{2} \\ \tan\dfrac{v}{2} = \sqrt{\dfrac{e+1}{e-1}}\tanh\dfrac{H}{2} \end{cases}$$

- So far, please note the beautiful analogy between elliptic & hyperbolic cases by defining *E* & *H* using auxiliary circle & equilateral hyperbola, respectively.

- Now that we have identified all conversions between *H* & *v*, we can derive the hyperbolic equation analogous to Kepler's equation in elliptic orbit:

  Start from (3)

$$\cos v = \frac{\cosh H - e}{1 - e\cosh H}$$

$$\Rightarrow -\sin v dv = \frac{\sinh H(1-e^2)}{\left(1-e\cosh H\right)^2}dH$$

$$\Rightarrow \vdots$$

$$\Rightarrow dv = -\frac{a\sqrt{e^2-1}}{r}dH \qquad (6)$$

  Now, again from

$$h = r^2 \frac{dv}{dt}$$

$$\Rightarrow hdt = r^2 dv \qquad \Leftarrow (6)$$

$$\Rightarrow \int_T^t hdt = -a\sqrt{e^2-1}\int_0^H rdH \qquad \Leftarrow \begin{cases} h = \sqrt{\mu p} \\ (5) \end{cases}$$

$$\Rightarrow (t-T) = \frac{-a\sqrt{e^2-1}}{\sqrt{\mu p}}\int_0^H a\left(1-e\cosh H\right)dH$$

$$\Rightarrow (t-T) = \sqrt{\frac{-a^3}{\mu}}\left(e\sinh H - H\right)$$

$$\Rightarrow M \equiv n(t-T) = e\sinh H - H \qquad \text{when } n = \sqrt{\frac{\mu}{-a^3}}$$

Generalizing for arbitrary position.

$$\frac{M-M_0}{n} = t - t_0 = \sqrt{\frac{-a^3}{\mu}}\left[\left(e\sinh H - H\right) - \left(e\sinh H_0 - H_0\right)\right]$$

2.2.4. (To be filled)

2.2.5. Solutions of Kepler's Equation

- One of the most common techniques is the Newton's iteration

  We like to solve a function

  $$f(y) = 0 \text{ for } y$$

  - Let's assume

    $$y = x + \delta$$

  - Then,

$$0 = f(y) = f(x+\delta) \approx f(x) + f'(x)\delta + \text{H.O.T}$$
$$\Rightarrow \delta \approx -\frac{f(x)}{f'(x)}$$

- $y_n = x_n + \delta_n$ improves our estimate
- Terminate if $\delta_n < \varepsilon$ (=desired tolerance)

- Applicable to elliptic/parabolic/hyperbolic cases
  - With modern computational power, the number of iteration should not be a problem in most cases
  - Refer to textbook if an efficient guess is necessary
    - Need to evaluate for multiple conditions
    - Some combinations of (M, e) requires a huge number of iterations (Refer to 3-D diagram in textbook)
  - In the parabolic case, we can even solve analytically by Cardano's formula
    - Barke's solution is just one way

2.2.6. Summary and Related Formulae

- Flight Path Angle

$$\phi_{fpa} = \phi_{fpa}(\nu, E, B, \text{ or } H)$$

- Start from

$$r\dot{\nu} = v\cos\phi_{fpa} \qquad \text{(tangential component)}$$

  - $$\Rightarrow \cos\phi_{fpa} = \frac{r\dot{\nu}}{v} \quad \Leftarrow \begin{cases} \dot{\nu} = \dfrac{h}{r^2} = \dfrac{\sqrt{\mu p}}{r^2}, \quad r = \dfrac{a(1-e^2)}{1+e\cos\nu} \\ v = \sqrt{\dfrac{2\mu}{r} - \dfrac{\mu}{a}} \end{cases}$$

$$\Rightarrow \cos\phi_{fpa} = \frac{1+e\cos\nu}{\sqrt{1+2e\cos\nu+e^2}} \tag{7}$$

$$\dot{r} = v\sin\phi_{fpa} \qquad \text{(radial component)}$$

■ $$\Rightarrow \sin\phi_{fpa} = \frac{\dot{r}}{v} \qquad \Leftarrow \begin{cases} r = \dfrac{a(1-e^2)}{1+e\cos\nu} \\ \dot{r} = \dfrac{r\dot{\nu}e\sin\nu}{1+e\cos\nu} \\ \dot{\nu} = \dfrac{h}{r^2} = \dfrac{\sqrt{\mu p}}{r^2} \\ v = \sqrt{\dfrac{2\mu}{r} - \dfrac{\mu}{a}} \end{cases}$$

$$\Rightarrow \sin\phi_{fpa} = \frac{e\sin\nu}{\sqrt{1+2e\cos\nu+e^2}} \tag{8}$$

■ (7) & (8) valid for any conic sections

- For the elliptic orbit, we can derive

$$\begin{cases} \cos\phi_{fpa} = \sqrt{\dfrac{1-e^2}{1-e^2\cos^2 E}} \\ \sin\phi_{fpa} = \dfrac{e\sin E}{\sqrt{1-e^2\cos^2 E}} \end{cases} \quad \Leftarrow \text{ simply use the relation between } (S_\nu, C_\nu) \Leftrightarrow (S_E, C_E)$$

- For the parabolic orbit,

$$\phi_{par} = \frac{\nu}{2}$$

- For the hyperbolic orbit,

$$\begin{cases} \sin\phi_{fpa} = \dfrac{-e\sinh H}{\sqrt{e^2\cosh^2 H - 1}} \\ \cos\phi_{fpa} = \sqrt{\dfrac{e^2-1}{e^2\cosh^2 H - 1}} \end{cases}$$

2.3. Kepler's Problem

- Kepler's Equation captures the time to travel between two known points on any type of orbit
- Kepler's Problem (Propagation)
  - Determines the location of a satellite in orbit after a certain amount of time
  - Find a satellite's future location, given the last known position/velocity vectors at a particular time

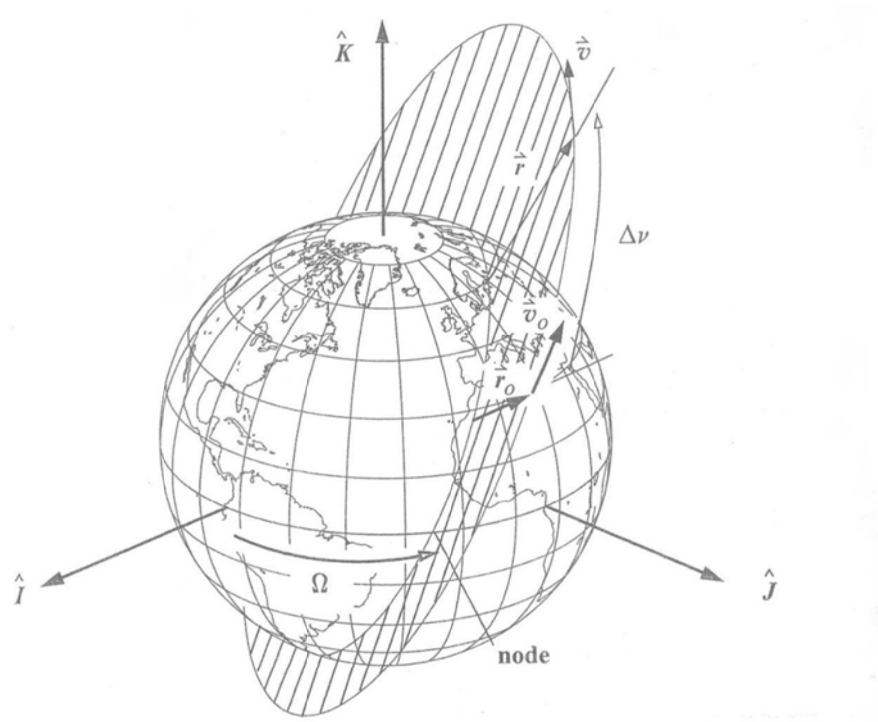


Figure 2-13. **Geometry for Kepler's Problem.** The concept behind Kepler's problem is the propagation of the satellite's state vector from an initial epoch to a future time.

2.3.1. Solution Techniques

2.3.1.1. Classical Orbital Elements

- A way to solve Kepler's Problem involves COE (Classical Orbital Elements)
  - a : SemimajorAxis
  - e : Eccentricity
  - i: Inclination
  - Ω: Right Ascention (Longitude) of the Ascending Node
  - ω : Argument of Periapsis
  - ν : True Anomaly
  - ➢ In the time between successive positions in the orbit, only ν changes

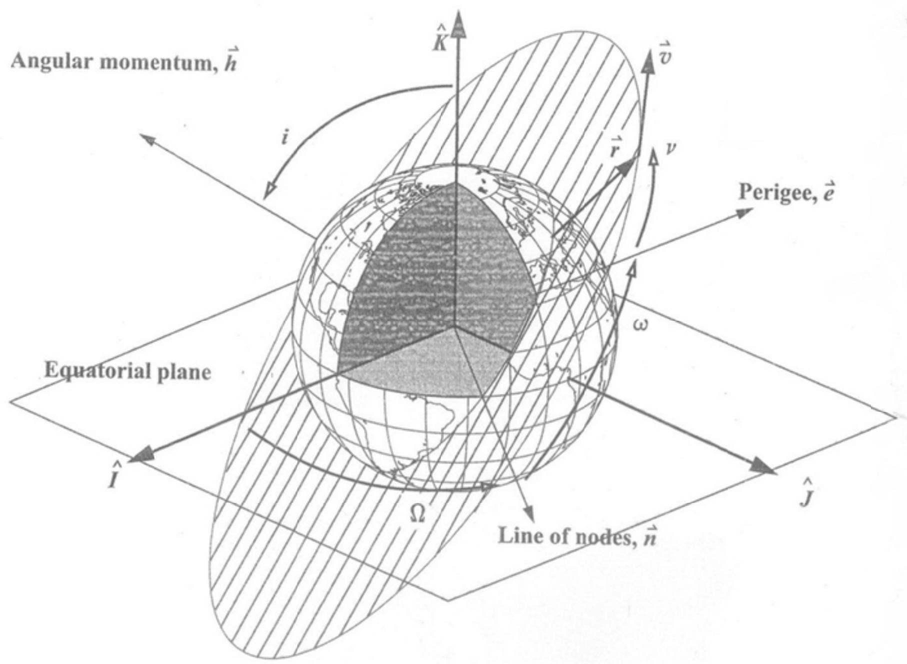


Figure 2-16. Classical Orbital Elements. The six classical orbital elements are the *semimajor axis, a*; *eccentricity, e*; *inclination, i*; *right ascension of the ascending node*, Ω—often referred to as simply the *node*; *argument of perigee*, ω; and *true anomaly*, ν. I haven't shown scale and shape elements (*a* and *e*) because I've introduced them in Chap. 1.

- Algorithm
  - $(\vec{r}_0,\vec{v}_0) \Rightarrow (a,e,i,\Omega,\omega,\nu_0)$
  - $$\begin{cases} M_0 = E_0 - e\sin E_0 & e<1 \\ M_0 = B_0 + \frac{1}{3}B_0^3 & e=1 \\ M_0 = e\sinh H_0 - H_0 & e>1 \end{cases}$$
  - $$\begin{cases} M = M_0 + n\Delta t \\ \quad \begin{cases} M,e \Rightarrow E & e<1 \\ \Delta t, p \Rightarrow B & e=1 \\ M,e \Rightarrow H & e>1 \end{cases} \end{cases}$$
  - $E,B,H \Rightarrow \nu$
  - $(a,e,i,\Omega,\omega,\nu) \Rightarrow (\vec{r},\vec{v})$

2.3.1.2. Classical Formulas Using *f* & *g* Functions

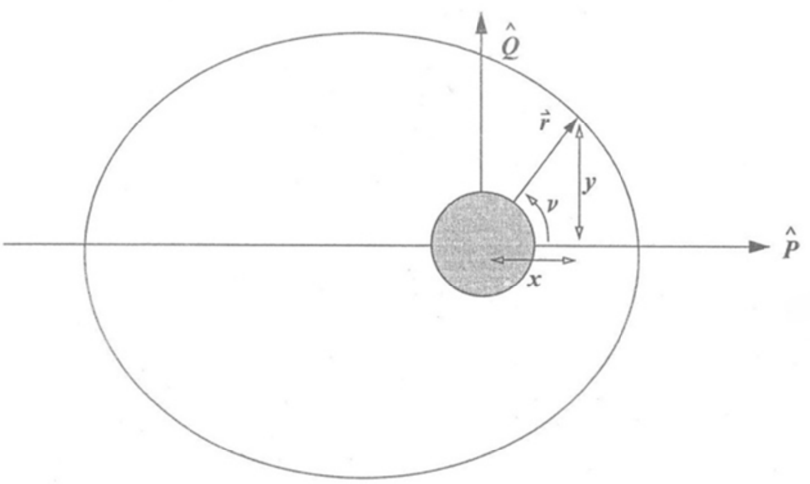


Figure 2-14. Geometry for the Perifocal Coordinate System. We can break a satellite's position into two components (x, y) in the orbital plane.

- Algorithm

  - In the perifocal coordinate system, position/velocity vectors can be expressed as

$$\vec{r} = x\hat{P} + y\hat{Q}$$
$$\vec{v} = \dot{x}\hat{P} + \dot{y}\hat{Q} \tag{A1}$$

  - Also assume $(\vec{r},\vec{v})$ as a linear combination of $(\vec{r}_0,\vec{v}_0)$

$$\vec{r} = f\vec{r}_0 + g\vec{v}_0$$
$$\vec{v} = \dot{f}\vec{r}_0 + \dot{g}\vec{v}_0 \tag{A2}$$

  - $$\begin{aligned}\vec{r}\times\vec{v}_0 &= \left(f\vec{r}_0 + g\vec{v}_0\right)\times\vec{v}_0 \qquad \Leftarrow (\text{A2}) \\ &= f\left(\vec{r}_0\times\vec{v}_0\right) + g\left(\vec{v}_0\times\vec{v}_0\right)\end{aligned}$$

$$\Rightarrow \vec{r}\times\vec{v}_0 = fh\hat{W} \tag{B1}$$

$$\vec{r}\times\vec{v}_0 = \begin{vmatrix}\hat{P} & \hat{Q} & \hat{W} \\ x & y & 0 \\ \dot{x}_0 & \dot{y}_0 & 0\end{vmatrix} = \left(x\dot{y}_0 - \dot{x}_0 y\right)\hat{W} \tag{B2}$$

  - (B1) = (B2)

$$\Rightarrow x\dot{y}_0 - \dot{x}_0 y = fh$$

$$\Rightarrow f = \frac{x\dot{y}_0 - \dot{x}_0 y}{h} \tag{D1}$$

$$\Rightarrow \dot{f} = \frac{\dot{x}\dot{y}_0 - \dot{x}_0\dot{y}}{h} \tag{D2}$$

  - $$\begin{aligned}\vec{r}_0\times\vec{r} &= \vec{r}_0\times\left(f\vec{r}_0 + g\vec{v}_0\right) \qquad \Leftarrow (\text{A2}) \\ &= f\left(\vec{r}_0\times\vec{r}_0\right) + g\left(\vec{r}_0\times\vec{v}_0\right)\end{aligned}$$

$$\Rightarrow \vec{r}_0\times\vec{r} = gh\hat{W} \tag{E1}$$

$$\vec{r}_0 \times \vec{r} = \begin{vmatrix} \hat{P} & \hat{Q} & \hat{W} \\ x_0 & y_0 & 0 \\ x & y & 0 \end{vmatrix} = \left(x_0 y - x y_0 \right) \hat{W} \tag{E2}$$

- (E1) = (E2)

  $\Rightarrow x_0 y - x y_0 = gh$

  $$\Rightarrow g = \frac{x_0 y - x y_0}{h} \tag{G1}$$

  $$\Rightarrow \dot{g} = \frac{x_0 \dot{y} - \dot{x} y_0}{h} \tag{G2}$$

(D1)~(D2) and (G1)~(G2) determine $\left(\vec{r}, \vec{v} \right)$ by (A2)

- We can check the accuracy of $\left(f, g \right)$

$$\begin{aligned} \vec{h} &= \vec{r} \times \vec{v} \\ &= \left(f\vec{r}_0 + g\vec{v}_0 \right) \times \left(\dot{f}\vec{r}_0 + \dot{g}\vec{v}_0 \right) \\ &= \left(f\dot{g} - \dot{f}g \right) \vec{h} \end{aligned}$$

$\Leftrightarrow 1 = fg' - f'g$ to be checked for accuracy

- Now we can use (f, g) with various solutions of *PQW* components, depending on what information is available. To determine (f, g) we must find (x, y) components

  - ν is known

$$\begin{cases} x = r\cos\nu & y = r\sin\nu \\ \dot{x} = -\sqrt{\dfrac{\mu}{p}}\sin\nu & \dot{y} = \sqrt{\dfrac{\mu}{p}}(e + \cos\nu) \end{cases}$$

$$\Rightarrow \begin{cases} f = 1 - \dfrac{r}{p}(1 - \cos\Delta\nu) \\ g = \dfrac{r r_0 \sin\Delta\nu}{\sqrt{\mu p}} \end{cases}$$

$$\Rightarrow \begin{cases} \dot{f} = \sqrt{\dfrac{\mu}{p}} \tan\left(\dfrac{\Delta\nu}{2}\right)\left[\dfrac{1 - \cos\Delta\nu}{p} - \dfrac{1}{r} - \dfrac{1}{r_0}\right] \\ \dot{g} = 1 - \left(\dfrac{r_0}{p}\right)\left(1 - \cos\Delta\nu\right) \end{cases}$$

■ E is known

$$\begin{cases} x = a\left[\cos E - e\right] \qquad y = a \sin E \sqrt{1 - e^2} \\ \Downarrow \quad \dot{E} = \dfrac{1}{r}\sqrt{\dfrac{\mu}{a}} \quad \Leftarrow \begin{cases} M = E - e \sin E \\ \dot{M} = \dot{E}(1 - e\cos E) \\ \dot{E} = \dfrac{\dot{M}}{1 - e\cos E} = \sqrt{\dfrac{\mu}{a^3}} \dfrac{1}{1 - e\cos E} \\ \qquad\quad \Uparrow \qquad\qquad\qquad \Uparrow \\ M = n(t - t_0) \Rightarrow \dot{M} = n = \sqrt{\dfrac{\mu}{a^3}} \\ r = a(1 - e\cos E) \Rightarrow \dfrac{a}{r} = \dfrac{1}{1 - e\cos E} \end{cases} \\ \dot{x} = -\dfrac{\sqrt{\mu a}}{r} \sin E \qquad \dot{y} = \dfrac{\sqrt{\mu a(1 - e^2)}}{r} \cos E \end{cases}$$

$$\Rightarrow \begin{cases} f = 1 - \dfrac{a}{r_0}(1 - \cos\Delta E) \\ g = (t - t_0) - \sqrt{\dfrac{a^3}{\mu}}\left(\Delta E - \sin\Delta E\right) \end{cases}$$

$$\Rightarrow \begin{cases} \dot{f} = -\dfrac{\sin\Delta E \sqrt{\mu a}}{r_0 r} \\ \dot{g} = 1 - \dfrac{a}{r_0}\left(1 - \cos\Delta E\right) \end{cases}$$

■ Similar results hold when B or H is known (Refer to Textbook)

2.3.1.3 Series Forms of (f, g)

- The series approach permits us to find expressions (f, g), when we do not know all COEs

- Especially useful when we know only the position magnitudes

- We attempt to represent the position vector at $t = \tau$ by Taylor series about $\vec{r}(t = 0)$:

  - $$\vec{r}(\tau) = \vec{r}_0 + \dot{\vec{r}}_0 \tau + \frac{\ddot{\vec{r}}_0}{2!}\tau^2 + \frac{\dddot{\vec{r}}_0}{3!}\tau^3 + \cdots \tag{9}$$

  - Note that we need to have additional derivatives at $t = 0$

$$\left.\begin{aligned}
&\dot{\vec{r}}_0 = \vec{v}_0 \\
&\ddot{\vec{r}}_0 = -\frac{\mu}{r_0^3}\vec{r}_0 = -u_0\vec{r}_0 \quad \Leftarrow \quad u \equiv \frac{\mu}{r^3} \\
&\dddot{\vec{r}}_0 = -\dot{u}_0\vec{r}_0 - u_0\dot{\vec{r}}_0 \\
&\vec{r}_0^{(4)} = \left[-\ddot{u}_0 + u_0^2\right]\vec{r}_0 - 2\dot{u}_0\dot{\vec{r}}_0
\end{aligned}\right) \tag{10}$$

  - Introducing (10) into (9), and grouping with respect to $\left(\vec{r}_0, \dot{\vec{r}}_0\right)$, we have

$$\vec{r} = f\vec{r}_0 + g\dot{\vec{r}}_0$$

$$\begin{cases}
f = 1 - \dfrac{u}{2}\tau^2 - \dfrac{\dot{u}}{6}\tau^3 - \dfrac{\ddot{u} - u^2}{24}\tau^4 - \cdots \\
g = \tau - \dfrac{u}{6}\tau^3 - \dfrac{\dot{u}}{12}\tau^4 - \dfrac{3\ddot{u} - u^2}{120}\tau^5 - \cdots
\end{cases}$$

- The simplest way to solve Kepler's Problem in modern era is to directly integrate 2-Body equations of motion with the given initial position/velocity

- Then, why use COE and/or (f, g)?
  - For analysis both qualitatively & quantitatively
  - Numerical integration is subject to numerical truncation/error, depending on pre-selected tolerance
    - May become serious for long-term integration
    - May lead to non-periodic solution due to numerical errors accumulated for sufficiently long time

2.4. Satellite State Representations

- We need 6 quantities to define what we call the 'state' of a satellite in space (Restricted Two-Body Problem)
  - State Vector = X: associated with position/velocity vectors
  - Element Set = Elset: Scalar magnitude and angular representation of the orbit

- Either set completely specify the two-body orbit and provide a complete set of initial conditions for solving an Initial Value Problem class of differential equation
  - State Vectors are reference to a particular coordinate frame
  - Elements Sets have a variety of forms
    - Classical Orbital Elements = Keplerian Elements = Two-Body Elements = $(a,e,i,\Omega,\omega,\nu)$
    - 2-Line Elements Sets
    - Canonical Elements
    - Delauney Elements
    - Poincare Elements
    - All are equivalent in essence, but different representations

2.4.1. Classical Orbital Elements

- When $(\vec{r}_0,\vec{v}_0)$ is known, we can calculate

$$\xi = \frac{v^2}{2} - \frac{\mu}{r} = \frac{v_0^2}{2} - \frac{\mu}{r_0}$$

$$\vec{h} = \vec{r} \times \vec{v} = \vec{r}_0 \times \vec{v}_0$$

■ Semimajor Axis (Scaler)

$$a = -\frac{\mu}{2\xi}, \qquad \Leftarrow \xi = -\frac{\mu}{2a}$$

$$a = \left(\frac{2}{r} - \frac{v^2}{\mu}\right)^{-1}, \quad \Leftarrow \xi = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$$

➢ Because $a \to \infty$ for parabola, p is sometimes used as the first orbital element ($p_{par} = 2r_p$)

■ Eccentricity (Shape)

◆ Let's start from the trajectory equation in vector form:

$$\dot{\vec{r}} \times \vec{h} = \mu\frac{\vec{r}}{r} + \vec{B}, \qquad \Leftarrow \vec{e} = \frac{\vec{B}}{\mu}$$

$$\Rightarrow \dot{\vec{r}} \times \vec{h} = \mu\frac{\vec{r}}{r} + \mu\vec{e}$$

$$\Rightarrow \vec{e} = \frac{\vec{v} \times \vec{h}}{\mu} - \frac{\vec{r}}{r}$$

$$\Rightarrow \vec{e} = \frac{\vec{v} \times (\vec{r} \times \vec{v})}{\mu} - \frac{\vec{r}}{r} \qquad \Rightarrow (\textit{lies in the orbital plane defined by} (\vec{r}, \vec{v}))$$

$$\Rightarrow \vec{e} = \frac{(\vec{v} \cdot \vec{v})\vec{r} - (\vec{r} \cdot \vec{v})\vec{v}}{\mu} - \frac{\vec{r}}{r}$$

$$\Rightarrow \vec{e} = \frac{\left(v^2 - \frac{\mu}{r}\right)\vec{r} - (\vec{r} \cdot \vec{v})\vec{v}}{\mu}$$

➢ The eccentricity vector turns out to always point to the periapsis

◆ In case we are interested in the magnitude of eccentricity only

$$p = a(1 - e^2), \qquad \Leftarrow p = \frac{h^2}{\mu}, a = -\frac{\mu}{2\xi}$$

$$\Rightarrow e = \sqrt{1 + \frac{2\xi h^2}{\mu^2}}$$

- Inclination (i)
    - Refers to the tilt of the orbital plane
    - Measured from z-axis to $\vec{h}$
    - $i \in [0,180]°$
        - $i = 0,180°$: equatorial orbits
        - $i \in (0,90)°$: travel with the rotation of the Earth (prograde)
        - $i \in (90,180)°$: oppose the rotation of the Earth (retrograde)
        - $i = 90°$: polar orbits

$$cos\, i = \frac{\hat{K} \cdot \vec{h}}{|\hat{K}||\vec{h}|}$$

- Right Ascension of the Ascending Node (RAAN, The Node, Ω)
    - Angle in the equatorial plane measured from X-axis to the location of the ascending node, which is the point on the equatorial plane at which the satellite crossed the equator from South to North
        - Descending Node
    - 'Line of Nodes' Connects both nodes to each other

$$\vec{n} = \hat{K} \times \vec{h}$$

- Equatorial orbits have no nodes
- $\Omega \in [0,360]°$

$$cos\, \Omega = \frac{\hat{I} \cdot \vec{n}}{|\hat{I}||\vec{n}|}$$

- If $n_j < 0$, then choose $360 - \Omega$

- The Argument of Periapsis (ω)

- Measured from the ascending node
- Locates the closest point of the orbit
- $\omega \in [0,360]°$

$$\cos\omega = \frac{\vec{n}\cdot\vec{e}}{|\vec{n}||\vec{e}|}$$

  - If $e_k < 0$, then choose $360 - \omega$

- True Anomaly (ν)
  - Determines the satellite's current position relative to the location of periapsis
  - $\nu \in [0,360]°$: requires quadrant check

$$\cos\nu = \frac{\vec{e}\cdot\vec{r}}{|\vec{e}||\vec{r}|}$$

    - If $\vec{r}\cdot\vec{v} < 0$, then choose $360 - \nu$
  - Undefined for circular orbits
- Time of Periapsis Passage (T)
  - Recall that

$$\tan\left(\frac{E}{2}\right) = \sqrt{\frac{1-e}{1+e}}\tan\left(\frac{\nu}{2}\right)$$

T isDetermined from Kepler's equation

$$M = n(t-T) = E - e\sin E$$

  - As we know $(\vec{r}_0,\vec{v}_0)$ completely, we can determine T
  - T usually gives a reference position in the orbit at one instant of time

2.4.1.1. A Method to solve Two-Body Problem

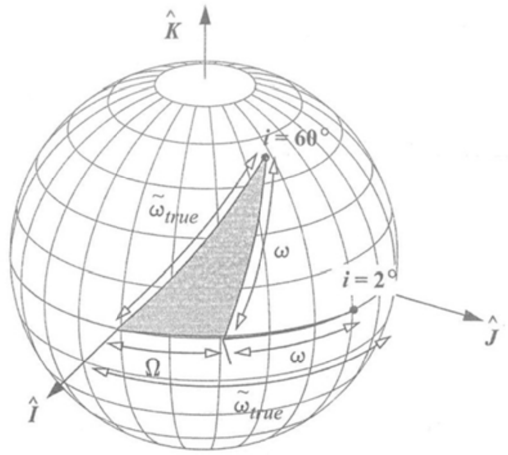
- Two-Body Problem is a 3 Degree of Freedom system
- A particular orbit is completely specified with 6 scalers

$$(\vec{r}(t_0),\dot{\vec{r}}(t_0)) \Leftrightarrow (a,e,i,\Omega,\omega,T)$$
$$\Leftrightarrow \quad \cdots$$

- How can we determine $(\vec{r}(t),\dot{\vec{r}}(t))$ at any given time t?
  - $M = n(t-T) = E - e\sin E \qquad \Rightarrow E(t)$
  - $\tan\left(\frac{\nu}{2}\right) = \sqrt{\frac{1+e}{1-e}}\tan\left(\frac{E}{2}\right) \qquad \Rightarrow \nu(t)$
  - $r = \frac{p}{1+e\cos\nu} = \frac{a(1-e^2)}{1+e\cos\nu} \qquad \Rightarrow r(t)$
  - $(r(t),\nu(t)) \Rightarrow \vec{r}(t)$
  - $v = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)} \qquad \Rightarrow |\dot{\vec{r}}(t)|$
  - $\cos\phi_{fpa} = \frac{1+e\cos\nu}{\sqrt{1+2e\cos\nu+e^2}} \qquad \Rightarrow \phi_{fpa}(t)$
  - $(|\dot{\vec{r}}(t)|,\phi_{fpa}(t)) \Rightarrow \dot{\vec{r}}(t)$

2.4.1.2. Special Cases

- A part of COE may not be defined for some special cases:
  - Elliptical Equatorial: $\tilde{\omega}_{true}$ (True Longitude of Periapsis)
  - Circular Inclined: u (Argument of Latitude)
  - Circular Equatorial: λ_{true} (True Longitude)



$i = 2°$, $e = 0.000\ 01$
$\Omega = 30°$, $\omega = 40°$
$\tilde{\omega}_{true} = 69.988\ 27°$

$i = 60°$, $e = 0.000\ 01$
$\Omega = 30°$, $\omega = 40°$
$\tilde{\omega}_{true} = 59.820\ 08°$

Figure 2-17. Representation of Longitude of Periapsis. Notice the effect inclination has on the values for the longitude of periapsis. The values of $(\Omega + \omega)$ and $\tilde{\omega}_{true}$ are close when the inclination is small but very different as the inclination increases.

- $\tilde{\omega}_{true}$ for Elliptical Equatorial (No Ascending Node)
  - Angle measured eastward from I-axis (vernal equinox) to the $\vec{e}$ vector (periapsis)
  - $\tilde{\omega}_{true} = \Omega + \omega$

$$\cos \tilde{\omega}_{true} = \frac{\hat{I} \cdot \vec{e}}{|\hat{I}||\vec{e}|}$$

  - If $e_j < 0$, then choose $360 - \tilde{\omega}_{true}$

- u for Circular Inclinded (No Periapsis)
  - Angle measured between the ascending node and the satellite's position vector

$$\cos u = \frac{\vec{n} \cdot \vec{r}}{|\vec{n}||\vec{r}|}$$

  - If $r_k < 0$, then choose $360 - u$

- λ_{true} for Circular Equatorial (No Ascending Node)
  - Angle measured eastward from I-axis (vernal equinox) to the position vector

$$\cos\lambda_{true} = \frac{\hat{I}\cdot\vec{r}}{|\hat{I}||\vec{r}|}$$

    - If $r_j < 0$, then choose $360 - \lambda_{true}$

2.5. Orbital Elements from $(\vec{r},\vec{v})$

- The conversion $(\vec{r},\vec{v}) \Leftrightarrow \left(a, e, \iota, \Omega, \omega, \nu(T)\right)$ is one of the most common problems in Astrodynamics.
  - Observations of a satellite are usually reduced to $(\vec{r},\vec{v})$
  - Knowledge of Orbital Elements makes the analysis much easier

- Algorithm $(\vec{r},\vec{v}) \Rightarrow \left(a, e, \iota, \Omega, \omega, \nu(T)\right)$

$$\vec{h} = \vec{r} \times \vec{v}, \quad h = \left|\vec{h}\right|$$
$$\vec{n} = \hat{K} \times \vec{h}$$

i. $$\vec{e} = \frac{\left(v^2 - \frac{\mu}{r}\right)\vec{r} - (\vec{r}\cdot\vec{v})\vec{v}}{\mu} \Rightarrow \quad e = |\vec{e}| \quad \text{or} \quad e = \sqrt{1 + \frac{2\xi h^2}{\mu^2}}$$

$$\xi = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$$

ii. $$\Rightarrow \begin{cases} e \neq 1.0 & a = -\dfrac{\mu}{2\xi}, \qquad p = a\left(1-e^2\right) \\ e = 1.0 & p = \dfrac{h^2}{\mu}, \qquad a \to \infty \end{cases}$$

iii. $$\cos\iota = \frac{h_K}{\left|\vec{h}\right|}$$

iv. $$\cos\Omega = \frac{n_I}{|\vec{n}|}, \quad \text{if } \mathrm{n}_J < 0, \quad \text{take } 360 - \Omega$$

v. $$\cos\omega = \frac{\vec{n}\cdot\vec{e}}{|\vec{n}||\vec{e}|}, \quad \text{if } \; e_K < 0, \quad \text{take } 360 - \omega$$

vi. $$\cos\nu = \frac{\vec{e}\cdot\vec{r}}{|\vec{e}||\vec{r}|}, \quad \text{if } \; \vec{r}\cdot\vec{v} < 0, \quad \text{take } 360 - \nu$$

Refer to Lecture Notes for special cases

2.6. $(\vec{r}, \vec{v})$ from Orbital Elements

- First determine $(\vec{r}, \vec{v})$ in the perifocal coordinate system

  i. Magnitude of position vector $r = \dfrac{p}{1 + e\cos\nu}$

  ii. $\vec{r}$ in perifocal coordinates

$$\vec{r}_{PQW} = \begin{bmatrix} r\cos\nu \\ r\sin\nu \\ 0 \end{bmatrix} = \begin{bmatrix} \dfrac{p\cos\nu}{1+e\cos\nu} \\ \dfrac{p\sin\nu}{1+e\cos\nu} \\ 0 \end{bmatrix}$$

  iii. $\vec{v}$ in perifocal coordinates

$$\vec{v}_{PQW} = \frac{d}{dt}\vec{r}_{PQW} = \begin{bmatrix} \dot{r}\cos\nu - r\dot{\nu}\sin\nu \\ \dot{r}\sin\nu + r\dot{\nu}\cos\nu \\ 0 \end{bmatrix}$$

$$\Rightarrow \vec{v}_{PQW} = \begin{bmatrix} -\sqrt{\dfrac{\mu}{p}}\sin\nu \\ \sqrt{\dfrac{\mu}{p}}\left(e + \cos\nu\right) \\ 0 \end{bmatrix}$$

- Then, transform into IJK frame using rotation matrix

  i. $\vec{r}_{IJK} = R_3\left(-\Omega\right) R_1\left(-\iota\right) R_3\left(-\omega\right) \vec{r}_{PQW}$

  ii. $\vec{v}_{IJK} = R_3\left(-\Omega\right) R_1\left(-\iota\right) R_3\left(-\omega\right) \vec{v}_{PQW}$

$$R_1(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & C_\phi & S_\phi \\ 0 & -S_\phi & C_\phi \end{bmatrix} \qquad \mathrm{R}_2(\theta) = \begin{bmatrix} C_\theta & 0 & -S_\theta \\ 0 & 1 & 0 \\ S_\theta & 0 & C_\theta \end{bmatrix} \qquad \mathrm{R}_3(\psi) = \begin{bmatrix} C_\psi & S_\psi & 0 \\ -S_\psi & C_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2.7. Application: Ground Tracks

- Ground Tracks are the locus of points formed by the points on the Earth directly
- Directly display a satellite's position as it travels through its orbit
- The change in longitude measured from the initial starting position, in one revolution gives us the period:

$$P = \frac{360^\circ - \Delta\lambda}{15^\circ / \text{sidereal hr}}$$

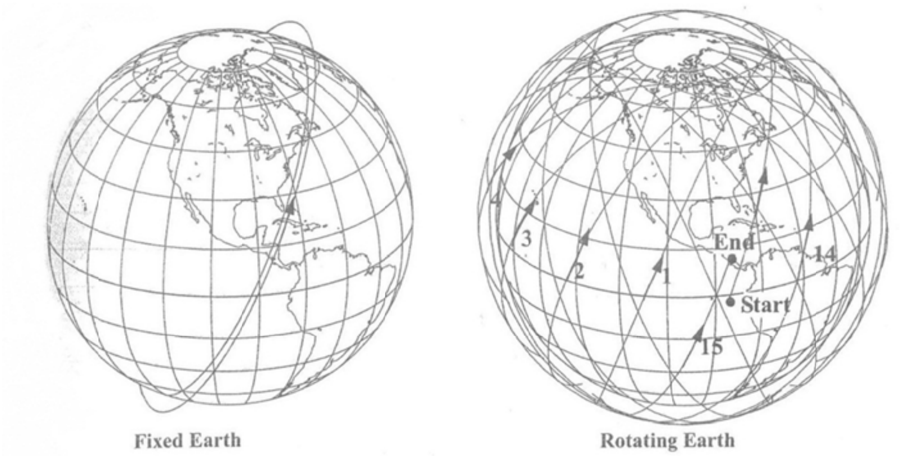


Figure 2-20. **Three-D Groundtracks.** With the Earth fixed, the satellite's orbital plane is visible, but this figure doesn't represent the satellite as it travels through time. In reality, the Earth moves under the "fixed" orbital plane. When a groundtrack is displayed over the positions it would occupy over time, the result is a "ball-of-yarn." Although this view *does* show the true positions, it's not particularly useful because it contains so much information. I've numbered each revolution for reference.

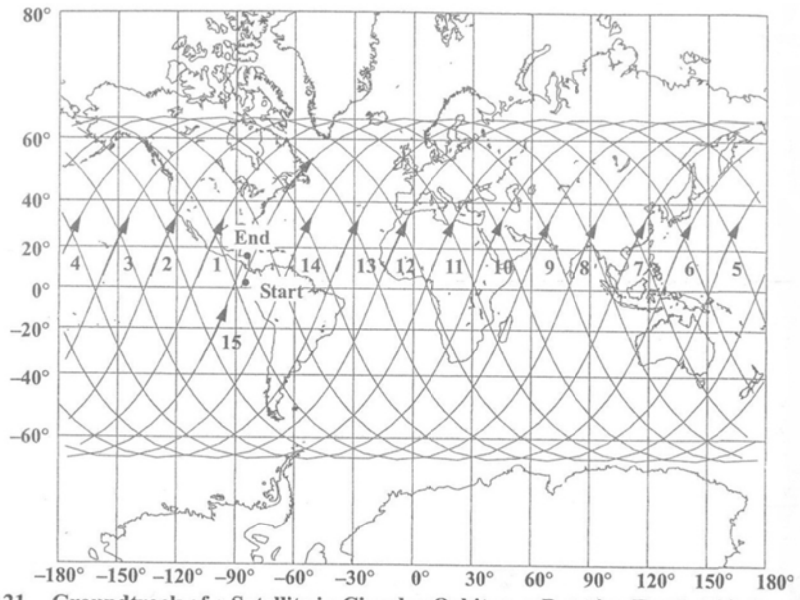


Figure 2-21. Groundtrack of a Satellite in Circular Orbit on a Rotating Earth. This form of the two-dimensional groundtrack is very useful because it shows the actual locations over a rotating Earth. Also notice the mismatch in the starting and ending points after 15 revolutions.

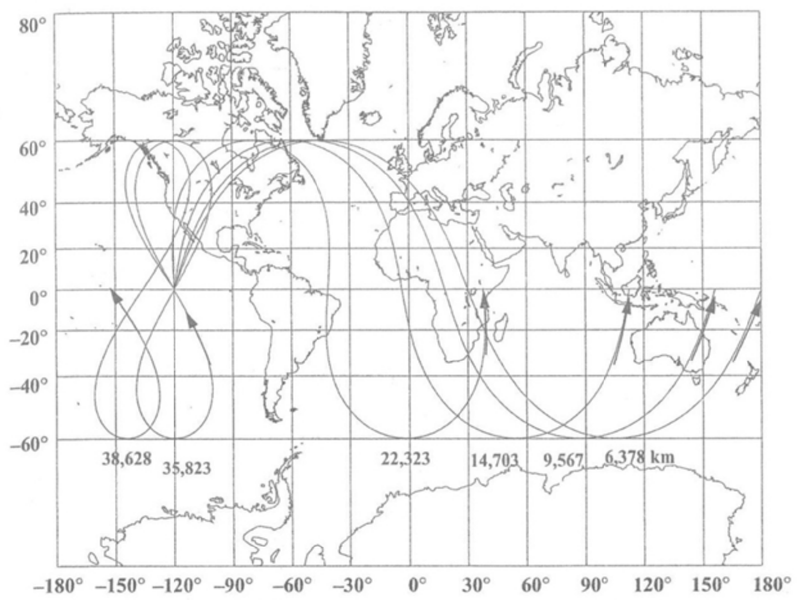


Figure 2-22. Effects of Increasing Altitude for Groundtracks on a Rotating Earth. Notice how the span of the groundtrack shrinks as the altitude increases. All groundtracks shown are for circular orbits. Altitudes are given in kilometers above the Earth's surface.

2.8. Application: Find Time of Flight

- We often need to determine Time of Flight between two locations on an orbit

- It is slightly different from Kepler's problem, because we know two position vectors
  - Need to find (a, p) such that Kepler's equation yield $\Delta t = t - t_0$
    - Equating (2-63) and (2-64)

$$\begin{cases} f = 1 - \left(\dfrac{r}{p}\right)(1 - COS(\Delta\nu)) & (2-63) \\ f = 1 - \dfrac{a}{r_o}(1 - COS(\Delta E)) & (2-64) \end{cases}$$

$$\Rightarrow \quad a = \frac{rr_o}{p}\frac{\left(1 - COS(\Delta\nu)\right)}{\left(1 - COS(\Delta E)\right)}$$

$$\begin{cases} \dot{f} = \sqrt{\dfrac{\mu}{p}} TAN\left(\dfrac{\Delta\nu}{2}\right)\left(\dfrac{1 - COS(\Delta\nu)}{p} - \dfrac{1}{r} - \dfrac{1}{r_o}\right) & (2-63) \\ \dot{f} = \dfrac{-SIN(\Delta E)\sqrt{\mu a}}{r_o r} & (2-64) \end{cases}$$

$$\Rightarrow \quad p = \frac{\left[1 - COS\left(\Delta\nu\right)\right] r_o r}{r_o + r - 2\sqrt{r_o r} COS\left(\dfrac{\Delta\nu}{2}\right) COS\left(\dfrac{\Delta E}{2}\right)}$$

Determine p first, and then a

  - Once we know (a, p), we can use (2-63) to determine (f, g). Then,

$$g = (t - t_o) - \sqrt{\frac{a^3}{\mu}}(\Delta E - SIN(\Delta E)) \qquad (2-64)$$

$$\Rightarrow (t - t_o) = g + \sqrt{\frac{a^3}{\mu}}(\Delta E - SIN(\Delta E))$$